

Xingyu Chen

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Education

University of Chinese Academy of Sciences (UCAS) | Sept. 2023 – Jun. 2027 (Expected)

B.Eng. in Artificial Intelligence (GPA 3.6/4)

Core Courses: Principles of Artificial Intelligence, Pattern Recognition and Machine Learning, Decision Intelligence for Complex Systems, Natural Language Processing, Introduction to Computer Science, C Programming and Design, Data Structures and Algorithms, Discrete Mathematics, Probability and Statistics.

University of California, Berkeley (UC Berkeley) | Jan. 2026 – May 2026

Visiting Student, UC Berkeley Extension

Core Courses: Mathematical and Graphical Foundations of Probability, Principles & Techniques of Data Science

Research Interests

I am interested in machine learning, focusing on natural language processing, sequence and generative modeling, and representation learning for complex, structured data to develop scalable algorithms with robust generalization. Building on these computational foundations, I aim to apply AI for Science to understand and engineer complex biological systems, spanning biological sequence and structure modeling, AI-driven drug discovery, multi-omics data integration, and biological dynamics.

Publications

Wang, L., Liu, Z., Tang, X., **Chen, X.** *et al.* FoldNav: Protein Structure Prediction as a Navigator for Masked Diffusion Protein Design. *Nat Mach Intell* (Under Review).

Research Experience

Machine Learning for Cellular Cryo-EM Denoising & Feature Classification | May 2026 – Present

Undergraduate Researcher | Supervisor: Prof. Bronwyn Lucas, UC Berkeley

- Constructed an automated 2D Template Matching (2DTM) refinement pipeline in Leopard-EM; optimized particle coordinate stacks via spatial filtering and fine-grained angular pose search.
- Developed algorithms for denoising and contrast enhancement of low-SNR cryo-EM images to identify ambiguous particle candidates under extreme noise conditions.
- Refined 3D pose estimation and orientation assignment for candidate particles, validating structural consistency using geodesic error metrics and 3D reconstructions.

Generate Specific Protein Sequences Based on the Generative Model | Oct. 2025 – July 2026

Undergraduate Researcher | Supervisor: Prof. Shu Wu, Institute of Automation, Chinese Academy of Sciences

- Developed a training-free framework that utilizes protein structure predictors (PSPs) as active navigators to steer diffusion-based sequence generation toward the manifold of foldable protein sequences.
- Engineered “Folding-Navigated Decoding” with periodic structural confidence injection and rejection sampling based on pLDDT, steric clashes, and stereochemical priors.
- Applied motif-folding LDDT guidance to motif regions by comparing natural and generated motif structures, and introduced dynamic re-masking for adaptive local resampling of low-pLDDT non-motif residues.
- Achieved improvements over baselines of 44.97% in geometric quality, 3.56% in foldability for unconditional design, 18.18% in motif-scaffolding pass rate, and 7.68% in inverse folding self-consistency.
- Co-authored a paper to summarize research findings, which is under review at *Nature Machine Intelligence*.

Analysis of Gut Microbiota Homeostasis Using Large Static Models | July 2025 – Aug. 2025

Member | Supervisor: Prof. Bin Sheng, UCAS-SJTU Joint Research Exchange Program

- Integrated human and murine pre-/post-intervention metagenomic data to construct phylogenetic trees; employed Graph Neural Networks (GNNs) to generate evolutionary-aware species embeddings for downstream tasks.
- Modeled cross-species intervention response patterns via transfer learning and feature alignment, enabling prediction of post-intervention human microbiota dynamics.
- Predicted host phenotypes from gut microbial traits (TCGs), quantifying associations with health outcomes and enabling indirect evaluation of drug efficacy.
- Migrated PopPhy-CNN from TensorFlow to PyTorch, improving architecture and data flow; implemented an MLP-based pipeline to validate the “representation–perturbation–comparison” workflow.

Academic Experience

Selection of NIPT Time Points and Fetal Abnormality Determination under Multi-factor Influence | Sept. 2025

Team Member | *Contemporary Undergraduate Mathematical Contest in Modeling (CUMCM)*

- Conducted feature screening by integrating Pearson correlation, XGBoost, causal inference, correlation ranking, and collinearity analysis; identified gestational age, BMI, and maternal age as key factors influencing Y-chromosome concentration.
- Built generalized additive model ($R^2=0.2176$) and random forest ($R^2=0.8654$), selecting GAM for interpretability and biological rationality; bootstrap validation yielded a pseudo- R^2 95% CI of [0.2059, 0.3038] and SD = 0.0243.
- Applied Nelder–Mead and simulated annealing algorithms to optimize NIPT timing, extending stratification from BMI alone to the two-dimensional BMI–age space and reducing CV from 34.38% to 17.48% at $\sigma=0.1$.
- Developed an XGBoost classifier for female fetus abnormality detection, achieving Recall=0.923 and F3-score=0.5314; performance remained robust under Gaussian noise (fluctuation range 0.029).

Prediction of Medal Outcomes for the 2028 Olympic Games | Jan. 2025

Team Leader | *Mathematical Contest in Modeling (MCM), COMAP*

- Built AFMP multivariate regression model with random search to predict medal counts of 77 countries; achieved prediction accuracy of over 60%.
- Designed “Medal Progress Index” to evaluate national performance trends; applied 3D K-means clustering to analyze potential first-time medal-winning countries.
- Evaluated host-country effect with Spearman correlation and second-order differencing, identifying an average 23.5% increase in host medal counts.
- Developed a regression-based scoring function to quantify the “Great Coaches” effect; proposed optimization strategies.
- Conducted Monte Carlo sensitivity analysis confirming stability and adaptability of the model.

Gomoku AI Project | Dec. 2024

Investigator | *Independent Project*

- Developed Gomoku program in C supporting six core modules, including human–machine/machine–machine play, first-move swap, game replay, and save/load functionality.
- Built dynamic chessboard with arrays and pointers; implemented recursive algorithm for win condition detection.
- Implemented forbidden move detection and optimized the scoring function via directional arrays and local simulation search.
- Employed structured arrays with binary read/write and timestamp-based archiving, enabling automated saving and replay for efficient game data management.

Lantern: Programmable Logic Framework Based on Recombinase and CRISPR Interference | Oct. 2024

Team Member | *International Genetically Engineered Machine Competition (iGEM)*

- Engineered programmable Boolean logic gates in *E. coli* using CRISPR interference (CRISPRi) and recombinase systems, enabling system-level control of cell behavior.
- Designed and validated 16 Boolean logic mappings, achieving proof-of-concept biomarker detection.
- Developed NLP-based circuit visualization tool to improve design readability and construction efficiency.
- Built mathematical models to assess promoter stability, estimate logic gate delays, and simulate CRISPRi suppression dynamics.

Honors and Awards

Honorable Mention Award (Top 15%), Mathematical Contest in Modeling (MCM), COMAP | May 2025

Silver Medal, International Genetically Engineered Machine Competition (iGEM) | Oct. 2024

Skills

Standardized Tests: TOEFL 5.0/6.0 (R5.0/L5.0/S4.5/W5.0); GRE 321

Programming: Python, C, C++, JavaScript

Data Analysis Tools: MATLAB, PyMOL, Mathematica, Origin, Stata, LINGO